For our second project for Semester 1, we worked on creating wind turbines that would be able to generate small amounts of electricity from gusts of wind available outside.

The turbine consists of five airfoils, most notably the root and the tip. The tip is the thinnest of all the airfoils and will be at the top of the turbine by design. It has a chord length of 0.03 meters, and an angle of attack of zero (0) degrees, a camber of 1.152%, and a thickness of 10.5%. The root is the thickest of all the airfoils and will be at the bottom, near the hub that connects all the turbines together. It has a chord length of 0.04 meters, an angle of attack of -5.037 degrees, a camber of 13.888%, and a thickness of 15%. There are three turbines in total, each separated by 120 degrees so that the wind turbine will stay balanced.

Some of the factors that can influence the performance of a wind turbine include the chord length, angle of attack, camber, thickness of the airfoils, air speed, how the wind turbine was constructed, the material used to construct the wind turbine, the height and the turbine's ability to start spinning because of the wind, errors during construction, the number of turbines and airfoils attached, and the wind turbines' strength and stability.

Because I do not have as much experience in engineering as of yet, I mainly followed the guidelines set by our teacher. In doing so, I was able to produce airfoils that adhered to the basic guidelines for this project. I used three turbines after getting inspired by the structure of the wind turbines along the Interstate 10 route to Los Angeles, California from Phoenix, Arizona, my hometown.

In most cases, from the data acquired from testing the airfoils through a wind tunnel, the airfoils experienced higher lift than drag. In the case of the root airfoil, a high speed allowed for a higher lift/drag (L/D) ratio. The root airfoil also had a higher L/D ratio at a higher angle of attack. The tip airfoil had a higher L/D ratio at a higher speed and lower angle of attack. The results of the wind tunnel testing were better than expected because I thought the drag force would be able to counter most of the lift, causing the wind turbines to be less efficient and generate less power.

Based on the wind tunnel testing results, the turbine performed as expected with the acquired indoor data. In both cases, as the speed of the wind increased, the power generated, and the lift and drag increased proportionally to the wind speed. However, with the acquired outdoor data, I expected the turbine to be able to generate more power because of the higher wind speeds outside. When I examined the data, the average power kept fluctuating and decreasing in the long-term, even though the wind speed stayed consistently high over the same time intervals these fluctuations occurred.

In the process of designing and building my turbine, the chopsticks I used to attach the airfoils to the hub kept breaking. Because I exerted a magnitude of force that was over the limits of what the chopsticks were able to handle, or I did not wait a long enough time for the glue to stick my turbine to the chopstick to dry, my chopsticks kept breaking. By waiting for longer periods of time, I was able to reinforce the turbine. While writing the code, because I had trouble translating my ideas into the syntax used in the Arduino IDE, I spent a longer time trying to understand how to use the correct syntax to allow the Arduino to perform in the way I wanted. Therefore, my code may not have included the complete logic behind the code.

While trying to print the data into the Arduino IDE Serial Monitor, I was not receiving any output. This was because my coding statements to print the outputs were in the wrong sequential order. Instead of putting the

coding statements at the end of my code, I relocated it to the void setup{} method so that the IDE would be able to print the previous data acquired easily.

Some potential sources of error in this project include the reinforcement of my turbines, the distance between airfoils, the tearing and subsequent patching of the paper wrapped around the airfoils, measuring the angle of attack of my turbines, balancing the blades on the hub, and converting floats into bytes to store into EEPROM easier.

With the reinforcement of my turbines, I tried to use the lowest amount of glue possible to stick both parts of the turbine together. With the distance between airfoils, I made sure to mark the point at which each airfoil should be placed so that there will be minimal interference between airfoils. Because the paper sometimes tore while testing, I would make sure to add the least amount of paper and glue, or tape to decrease any unwanted effects on the turbine.

While trying to measure the angle of attack and balance the blades on the hub, I made sure to balance the blades on the hub within a specific range of angles so as not to severely impact the angle of attack. While trying to convert floats to bytes for easier storage, I kept the float value until the point where we had to map the power into a byte for the EEPROM to store easily. In doing so, the significant figures are preserved until presenting a final solution.

Overall, I think the sources of error affected my data to a moderate degree because it compromised the structural integrity of my wind turbines.

If given a fresh opportunity, I would like to increase the number of blades I have so that the wind turbine is able to catch more wind. Furthermore, I would like to try experimenting with a vertical wind turbine to understand what parameters impact its ability to be scaled.